Started on	Tuesday, 17 March 2026, 9:39 AM
State	Finished
Completed on	Tuesday, 17 March 2026, 10:15 AM
Time taken	36 mins 21 secs
Grade	**0.00** out of 100.00

Question 1

Incorrect

Mark 0.00 out of 15.00

Write the methods with the following headers:

// Return the reversal of an integer, e.g., reverse(456) returns 654

public static int ***reverse****(int number)*

// Return true if number is a palindrome

public static boolean ***isPalindrome****(int number)*

Use the **reverse** method to implement **isPalindrome**. A number is a palindrome if its reversal is the same as itself. Write a test program that prompts the user to enter an integer and reports whether the integer is a palindrome.

For example:

Input	Result
466	Enter a postive integer: 466 466 is not a palindrome
7887	Enter a postive integer: 7887 7887 is a palindrome

Answer: (penalty regime: 0 %)

```
import java.util.Scanner;

public class Q1{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter a positive integer: ");
        int number = input.nextInt();
        System.out.println(number);


        if (isPalindrome(number))
          System.out.println(number + " is a palindrome");
        else
          System.out.println(number + " is not a palindrome");
    }
    public static boolean isPalindrome(int number){
        return number == reverse(number);
    }

    public static int reverse(int number) {
        int result = 0;
        while (number != 0){
            int remainder = number % 10 ;

            result = result * 10 + remainder;

            number = number / 10;
        }
        return result;
    }
}
```

	Input	Expected	Got	
✖	466	Enter a postive integer: 466 466 is not a palindrome	Enter a positive integer: 466 466 is not a palindrome	✖
✖	7887	Enter a postive integer: 7887 7887 is a palindrome	Enter a positive integer: 7887 7887 is a palindrome	✖

Some hidden test cases failed, too.

Your code must pass all tests to earn any marks. Try again.

Show differences

Show/hide question author's solution (Java)

```java
import java.util.Scanner;

public class Exercise06_03 {
  public static void main(String[] args) {
    Scanner input = new Scanner(System.in);
    System.out.print("Enter a postive integer: ");
    int number = input.nextInt();
    System.out.println(number);

    if (isPalindrome(number))
      System.out.println(number + " is a palindrome");
    else
      System.out.println(number + " is not a palindrome");
  }

  public static boolean isPalindrome(int number) {
    return number == reverse(number);
  }

  public static int reverse(int number) {
    int result = 0;
    while (number != 0) {
      int remainder = number % 10;

      result = result * 10 + remainder;

      number = number / 10;
    }

    return result;
  }
}

```

Incorrect

Marks for this submission: 0.00/15.00.

Question 2

Not answered

Mark 0.00 out of 15.00

Some Websites impose certain rules for passwords. Write a method that checks whether a string is a valid password. Suppose the password rules are as follows:

■ A password must have at least **eight** characters.

■ A password must contain **only** letters and digits.

■ A password must contain at least **two** digits.

Write a program that prompts the user to enter a password and displays "**Valid Password**" if the rules are followed, or "**Invalid Password**" otherwise.

For example:

Input	Result
password123	Enter a string for password: password123 Valid Password
14+codingJava	Enter a string for password: 14+codingJava Invalid Password

Answer: (penalty regime: 0 %)

```
1
```

Show/hide question author's solution (Java)

```java
public class Exercise06_18 {
  public static void main(String[] args) {
    // Prompt the user to enter a password
    java.util.Scanner input = new java.util.Scanner(System.in);
    System.out.print("Enter a string for password: ");
    String s = input.nextLine();
    System.out.println(s);

    if (isValidPassword(s)) {
      System.out.println("Valid Password");
    }
    else {
      System.out.println("Invalid Password");
    }
  }

  /** Check if a string is a valid password */
  public static boolean isValidPassword(String s) {
    // Only letters and digits?
    for (int i = 0; i < s.length(); i++) {
      if (!Character.isLetter(s.charAt(i)) &&
          !Character.isDigit(s.charAt(i)))
```

```
        return false;
    }

    // Check length
    if (s.length() < 8)
      return false;

    // Count the number of digits
    int count = 0;
    for (int i = 0; i < s.length(); i++) {
      if (Character.isDigit(s.charAt(i)))
        count++;
    }

    if (count >= 2)
      return true;
    else
      return false;
  }
}
```

Question 3

Not answered

Mark 0.00 out of 15.00

Write a program that reads student scores, gets the best score, and then assigns grades based on the following scheme:

Grade is A if score is ≥ best -10;

Grade is B if score is ≥ best -20;

Grade is C if score is ≥ best -30;

Grade is D if score is ≥ best -40;

Grade is F otherwise.

The program prompts the user to enter the **total number** of students, then prompts the user to enter all of the scores, and concludes by displaying the grades.

Hint: Use a single-dimensional array to store the scores of all students.

For example:

Input	Result
5 45 85 62 91 13	Enter number of students: 5 Enter 5 scores: 45.0 85.0 62.0 91.0 13.0 Student 0 score is 45.0 and grade is F Student 1 score is 85.0 and grade is A Student 2 score is 62.0 and grade is C Student 3 score is 91.0 and grade is A Student 4 score is 13.0 and grade is F
3 99 78 43	Enter number of students: 3 Enter 3 scores: 99.0 78.0 43.0 Student 0 score is 99.0 and grade is A Student 1 score is 78.0 and grade is C Student 2 score is 43.0 and grade is F

Answer: (penalty regime: 0 %)

```
1
```

▼ **Show/hide question author's solution (Java)**

```java
import java.util.Scanner;

public class Exercise07_01 {
  /** Main method */
  public static void main(String[] args) {
    // Create a Scanner
    Scanner input = new Scanner(System.in);

    // Get number of students
    System.out.print("Enter number of students: ");
```

```
    int numberOfStudents = input.nextInt();
    System.out.println(numberOfStudents);

    double[] scores = new double[numberOfStudents]; // Array scores
    double best = 0; // The best score

    // Read scores and find the best score
    System.out.print("Enter " + numberOfStudents + " scores: ");
    for (int i = 0; i < scores.length; i++) {
      scores[i] = input.nextDouble();
      System.out.print(scores[i] + " ");
      if (scores[i] > best)
        best = scores[i];
    }
    System.out.println();

    // Declare and initialize output string
    char grade; // The grade

    // Assign and display grades
    for (int i = 0; i < scores.length; i++) {
      if (scores[i] >= best - 10)
        grade = 'A';
      else if (scores[i] >= best - 20)
        grade = 'B';
      else if (scores[i] >= best - 30)
        grade = 'C';
      else if (scores[i] >= best - 40)
        grade = 'D';
      else
        grade = 'F';

      System.out.println("Student " + i + " score is " + scores[i] + " and grade is " + grade);
    }
  }
}

```

Question 4

Incorrect

Mark 0.00 out of 15.00

Write a method that finds the smallest element in an array of double values using the following header:

public static double min(double[] array)

Write a test program that prompts the user to enter 7 numbers, invokes this method to return the minimum value, and displays the **minimum** value.

For example:

Input	Result
15 7 124 56 235 2 68	Enter seven double numbers: 15.0 7.0 124.0 56.0 235.0 2.0 68.0 The minimal value is 2.0
102 543 8 301 56 8 92	Enter seven double numbers: 102.0 543.0 8.0 301.0 56.0 8.0 92.0 The minimal value is 8.0

Answer: (penalty regime: 0 %)

```
public class ExerciseQ4 {
  public static void main(String[] args) {
    double[] numbers = new double[7];

    java.util.Scanner input = new java.util.Scanner(System.in);
    System.out.print("Enter seven double numbers: ");

    for (int i = 0; i < numbers.length; i++){
      numbers[i] = input.nextDouble();
      System.out.print(numbers[i] + " ");
    }
      System.out.println();
      System.out.println("The minimal value is " + min(numbers));
  }
  public static double min(double[] list) {
    double min = list[0];
for (int i = 1; i < list.length; i++);
if (min > list[i])
        min = list[i];
    return min;
  }
}
```

Syntax Error(s)

```
ExerciseQ4.java:18: error: cannot find symbol
if (min > list[i])
              ^
  symbol:   variable i
  location: class ExerciseQ4
ExerciseQ4.java:19: error: cannot find symbol
        min = list[i];
                   ^
  symbol:   variable i
  location: class ExerciseQ4
2 errors
```

Show/hide question author's solution (Java)

```
public class Exercise07_09 {
  // Main method
  public static void main(String[] args) {
    double[] numbers = new double[7];

    java.util.Scanner input = new java.util.Scanner(System.in);
    System.out.print("Enter seven double numbers: ");

    for (int i = 0; i < numbers.length; i++){
      numbers[i] = input.nextDouble();
      System.out.print(numbers[i] + " ");
    }
```

```
13        System.out.println();
14        System.out.println("The minimal value is " + min(numbers));
15    }
16
17    public static double min(double[] list) {
18      double min = list[0];
19
20      for (int i = 1; i < list.length; i++)
21        if (min > list[i])
22          min = list[i];
23
24      return min;
25    }
26  }
27
```

Incorrect

Marks for this submission: 0.00/15.00.

Question 5

Not answered

Mark 0.00 out of 20.00

Suppose you need to write a program that grades multiple-choice tests. Assume there are eight students and ten questions, and the answers are stored in a two-dimensional array. Each row records a student's answers to the questions, as shown in the following array:

	0	1	2	3	4	5	6	7	8	9
Student 0	A	B	A	C	C	D	E	E	A	D
Student 1	D	B	A	B	C	A	E	E	A	D
Student 2	E	D	D	A	C	B	E	E	A	D
Student 3	C	B	A	E	D	C	E	E	A	D
Student 4	A	B	D	C	C	D	E	E	A	D
Student 5	B	B	E	C	C	D	E	E	A	D
Student 6	B	B	A	C	C	D	E	E	A	D
Student 7	E	B	E	C	C	D	E	E	A	D

The key is stored in a one-dimensional array:

	0	1	2	3	4	5	6	7	8	9
Key	D	B	D	C	C	D	A	E	A	D

Your program grades the test and displays the result. It compares each student's answers with the key, counts the number of correct answers, and displays it.

For example:

Input	Result
1	Student 0's correct count is 7 Student 1's correct count is 6 Student 2's correct count is 5 Student 3's correct count is 4 Student 4's correct count is 8 Student 5's correct count is 7 Student 6's correct count is 7 Student 7's correct count is 7

Answer: (penalty regime: 0 %)

```
1
```

▼ **Show/hide question author's solution (Java)**

```
1 ▾ public class GradeExam {
```

```
/** Main method */
public static void main(String[] args) {
// Students' answers to the questions
char[][] answers = {
{'A', 'B', 'A', 'C', 'C', 'D', 'E', 'E', 'A', 'D'},
{'D', 'B', 'A', 'B', 'C', 'A', 'E', 'E', 'A', 'D'},
{'E', 'D', 'D', 'A', 'C', 'B', 'E', 'E', 'A', 'D'},
{'C', 'B', 'A', 'E', 'D', 'C', 'E', 'E', 'A', 'D'},
{'A', 'B', 'D', 'C', 'C', 'D', 'E', 'E', 'A', 'D'},
{'B', 'B', 'E', 'C', 'C', 'D', 'E', 'E', 'A', 'D'},
{'B', 'B', 'A', 'C', 'C', 'D', 'E', 'E', 'A', 'D'},
{'E', 'B', 'E', 'C', 'C', 'D', 'E', 'E', 'A', 'D'}};

// Key to the questions
char[] keys = {'D', 'B', 'D', 'C', 'C', 'D', 'A', 'E', 'A', 'D'};

// Grade all answers
for (int i = 0; i < answers.length; i++) {
// Grade one student
int correctCount = 0;
for (int j = 0; j < answers[i].length; j++) {
if (answers[i][j] == keys[j])
correctCount++;
}

System.out.println("Student " + i + "'s correct count is " + correctCount);
}
}
}
```

Question 6

Not answered

Mark 0.00 out of 20.00

Write a method that returns the sum of all the elements in a specified column in a matrix using the following header:

public static double sumColumn(double[][] m, int columnIndex)

Write a test program that reads a 3-by-4 matrix and displays the sum of each column.

For example:

Input	Result
1.5 2 3 4 5.5 6 7 8 9.5 1 3 1	Enter a 3 by 4 matrix row by row: 1.5 2.0 3.0 4.0 5.5 6.0 7.0 8.0 9.5 1.0 3.0 1.0 Sum of the elements at column 0 is 16.5 Sum of the elements at column 1 is 9.0 Sum of the elements at column 2 is 13.0 Sum of the elements at column 3 is 13.0
3.75 5 3 7 7.15 1 4 8 9.12 7 6 1	Enter a 3 by 4 matrix row by row: 3.75 5.0 3.0 7.0 7.15 1.0 4.0 8.0 9.12 7.0 6.0 1.0 Sum of the elements at column 0 is 20.02 Sum of the elements at column 1 is 13.0 Sum of the elements at column 2 is 13.0 Sum of the elements at column 3 is 16.0

Answer: (penalty regime: 0 %)

```
1
```

Show/hide question author's solution (Java)

```java
import java.util.Scanner;

public class Exercise08_01 {
  public static void main(String[] args) {
    Scanner input = new Scanner(System.in);

    System.out.println("Enter a 3 by 4 matrix row by row: ");
    double[][] m = new double[3][4];

    for (int i = 0; i < m.length; i++){
      for (int j = 0; j < m[i].length; j++){
        m[i][j] = input.nextDouble();
        System.out.print(m[i][j] + " ");
      }
      System.out.println();
```

```
    }
    for (int j = 0; j < m[0].length; j++) {
      System.out.println("Sum of the elements at column " + j + " is " +
        sumColumn(m, j));
    }
  }

  public static double sumColumn(double[][] m, int columnIndex) {
    double total = 0;

    for (int i = 0; i < m.length; i++)
      total += m[i][columnIndex];

    return total;
  }
}
```